

TIØ4145 Corporate Finance

Exercise 1: Capital budgeting and valuation of bonds

Deadline: Monday September 1st, 2025, 12:00 PM – Grading: Passed/Failed

Your answers should be submitted as a PDF file to BlackBoard before the deadline, either written in MS Word, LaTeX or handwritten with a high-quality scan. You are also welcome to upload additional Excel computations if you find it necessary. Guidance to this exercise is given on Thursday August 28th, 8:15 – 10:00 in A32. Solutions to the problems will be posted on BlackBoard after the deadline.

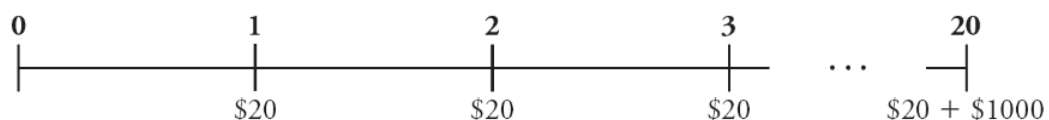
Relevant Curriculum

- Lecture notes from lecture given on August 25th.
- Berk and DeMarzo, Corporate Finance 5th edition: Chapters 6 and 8

Tasks

Chapter 6 – Valuing Bonds

1. Suppose a five-year, \$1000 bond with annual coupons has a price of \$1050 and a yield to maturity of 6%. What is the bond's coupon rate?
2. Assume that a bond will make payments every six months as shown on the following timeline (using six-month periods):



- a. What is the maturity of the bond (in years)?
- b. What is the coupon rate (in percent)?
- c. What is the face value?

3. Suppose the current zero-coupon yield curve for risk-free bonds is as follows:



Maturity (years)	1	2	3	4	5
YTM	4.28%	4.75%	4.89%	5.20%	5.45%

- What is the price per \$100 face value of a three-year, zero-coupon, risk-free bond?
 - What is the price per \$100 face value of a four-year, zero-coupon, risk-free bond?
 - What is the risk-free interest rate for a three-year maturity?
4. Suppose you purchase a 30-year, zero-coupon bond with a yield to maturity of 6%. You hold the bond for five years before selling it. The face value is \$100.
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- If the bond's yield to maturity is 6% when you sell it, what is the internal rate of return of your investment?
 - If the bond's yield to maturity is 7% when you sell it, what is the internal rate of return of your investment?
 - If the bond's yield to maturity is 5% when you sell it, what is the internal rate of return of your investment?
5. The following table summarizes the yield to maturity on several one-year, zero-coupon securities:

Security	Yield (%)
Treasury	2.940
AAA corporate	3.633
BBB corporate	4.259
B corporate	5.061

- What is the price (expressed as a percentage of the face value) of a one-year zero-coupon corporate bond with an AAA rating?
- What is the credit spread on AAA-rated corporate bonds?
- What is the spread on B-rated corporate bonds?
- How does the credit spread change with the bond rating? Why?

Exam Question – August 2015

Today ($t = 0$), the prevailing annualized interest rates for investments of different year horizons are as follows: $r_{0,1} = 10\%$; $r_{0,2} = 6\%$ and $r_{0,3} = 8\%$. ($r_{0,t}$ denotes the holding period interest rate from 0 to t .) These rates together comprise the yield curve. They also represent

the annualized rates of return that can be achieved on zero-coupon bonds of the corresponding maturity. You own a 3-year bond with face value 100000 NOK and 8% coupon paid annually. You are thinking of buying a car in two years.

What is the price of the bond today?

Chapter 8 – Fundamentals of Capital Budgeting

1.  Beryl's Iced Tea currently rents a bottling machine for \$51,000 per year, including all maintenance expenses. It is considering purchasing a machine instead, and is comparing two options:

- a. Purchase the machine it is currently renting for \$160,000. This machine will require \$21,000 per year in ongoing maintenance expenses.
- b. Purchase a new, more advanced machine for \$255,000. This machine will require \$18,000 per year in ongoing maintenance expenses and will lower bottling costs by \$11,000 per year. Also, \$37,000 will be spent upfront in training the new operators of the machine.

Suppose the appropriate discount rate is 8% per year and the machine is purchased today. Maintenance and bottling costs are paid at the end of each year, as is the rental of the machine. Assume also that the machines will be depreciated via the straight-line method over seven years and that they have a 10-year life with a negligible salvage value. The marginal corporate tax rate is 30%. Should Beryl's Iced Tea continue to rent, purchase its current machine, or purchase the advanced machine?

2. Arnold Inc. is considering a proposal to manufacture high-end protein bars used as food supplements by body builders. The project requires use of an existing warehouse, which the firm acquired three years ago for \$3 million and which it currently rents out for \$123,000. Rental rates are not expected to change going forward. In addition to using the warehouse, the project requires an upfront investment into machines and other equipment of \$1.3 million. This investment can be fully depreciated straight-line over the next 10 years for tax purposes. However, Arnold Inc. expects to terminate the project at the end of eight years and to sell the machines and equipment for \$496,000. Finally, the project requires an initial investment into net working capital equal to 10% of predicted first-year sales. Subsequently, net working capital is 10% of the predicted sales over the following year. Sales of protein bars are expected to be \$4.5 million in the first year and to stay constant for eight years. Total manufacturing costs and operating expenses (excluding depreciation) are 80% of sales, and profits are taxed at 30%.

- a. What are the free cash flows of the project?
- b. If the cost of capital is 15%, what is the NPV of the project?

3.  Bauer Industries is an automobile manufacturer. Management is currently evaluating a proposal to build a plant that will manufacture lightweight trucks. Bauer plans to use a cost of capital of 12% to evaluate this project. Based on extensive research, it has prepared the following incremental free cash flow projections (in millions of dollars):

	Year 0	Years 1-9	Year 10
Revenues		99.0	99.0
- Manufacturing expenses (other than depreciation)		-35.4	-35.4
- Marketing expenses		-9.5	-9.5
- Depreciation		-15.5	-15.5
= EBIT		38.6	38.6
- Taxes (34%)		-13.124	-13.124
= Unlevered net income		25.476	25.476
+ Depreciation		+15.5	+15.5
- Increases in net working capital		-5.1	-5.1
- Capital expenditures	-155.0		
+ Continuation Value			+12.1
= Free cash flow	-155.0	35.876	47.976

- a. For this base-case scenario, what is the NPV of the plant to manufacture lightweight trucks?
- b. Based on input from the marketing department, Bauer is uncertain about its revenue forecast. In particular, management would like to examine the sensitivity of the NPV to the revenue assumptions. What is the NPV of this project if revenues are 12% higher than forecast? What is the NPV if revenues are 12% lower than forecast?
- c. Rather than assuming that cash flows for this project are constant, management would like to explore the sensitivity of its analysis to possible growth in revenues and operating expenses. Specifically, management would like to assume that revenues, manufacturing expenses, and marketing expenses are as given in the table for year 1 and grow by 2% per year every year starting in year 2. Management also plans to assume that the initial capital expenditures (and therefore depreciation), additions to net working capital, and continuation value remain as initially specified in the table. What is the NPV of this project under these alternative assumptions? How does the NPV change if the revenues and operating expenses grow by 6% per year rather than by 2%?
- d. To examine the sensitivity of this project to the discount rate, management would like to compute the NPV for different discount rates. Create a graph, with the discount rate on the x-axis and the NPV on the y-axis, for discount rates ranging

from 5% to 30%. For what ranges of discount rates does the project have a positive NPV?

References

Berk, J. and DeMarzo, P. (2017). Corporate Finance, Fifth Edition, Pearson Education, Essex, England.